INCH-POUND

MIL-DTL-55302/111B
22 May 2006
SUPERSEDING
MIL-C-55302/111A
12 December 1979

DETAIL SPECIFICATION SHEET

# CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES: RECEPTACLE SOCKET CONTACTS, STRAIGHT-THRU, FOR MULTILAYERED PRINTED WIRING BOARDS (.100 SPACING)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-55302.

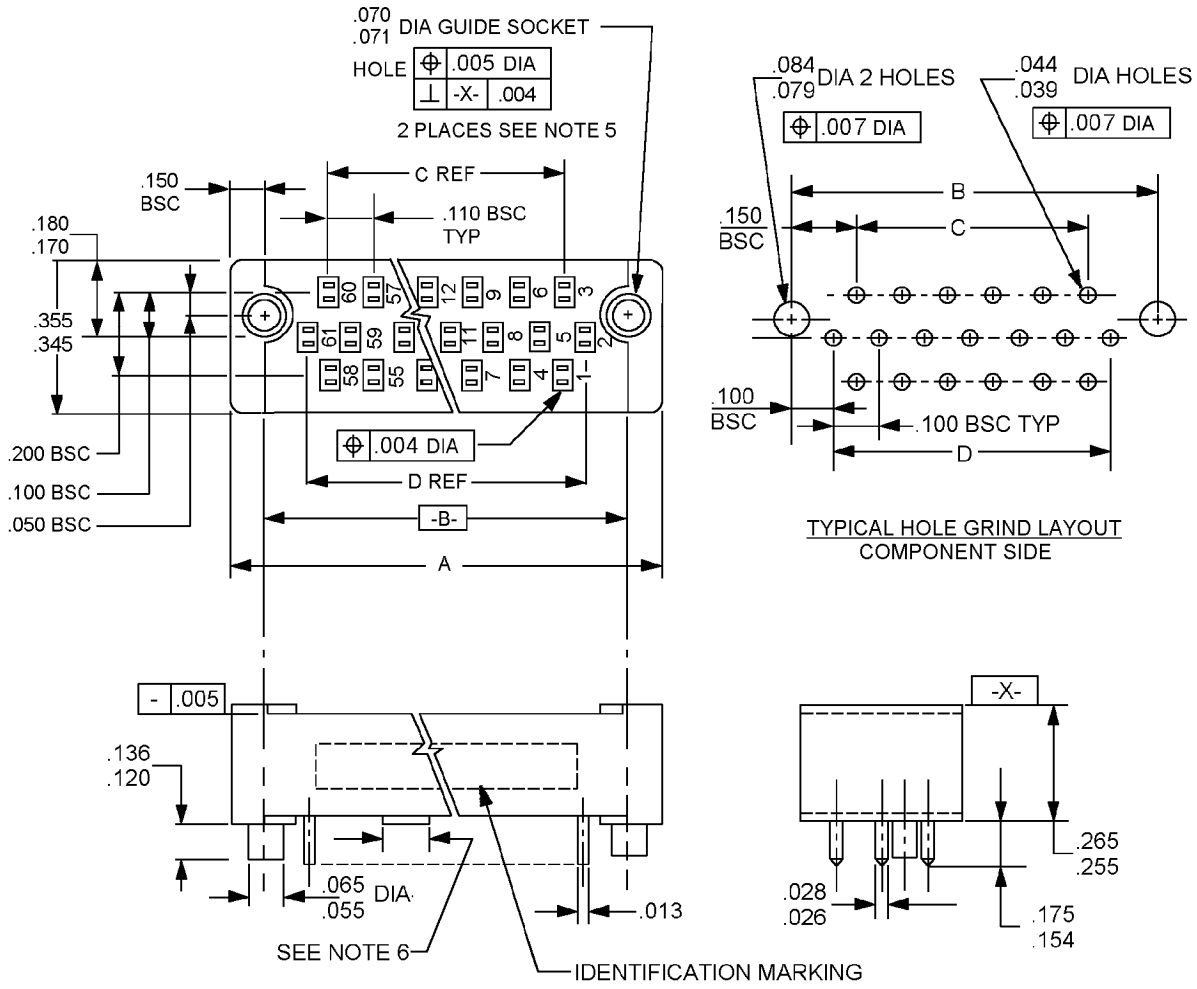



FIGURE 1. Connectors receptacle, .100 (2.54 mm) spacing.

AMSC N/A

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TABLE I. Dash numbers and dimensions for types II and III.

| Dash numbers | Number of contacts | A | B Basic | C Ref. | D Ref |
|---|---|---|---|---|---|
| 01 | 13 | .800 ±.005 | .600 | .300 | .400 |
| 02 | 25 | 1.200 ±.005 | 1.000 | .700 | .800 |
| 03 | 37 | 1.600 ±.005 | 1.400 | 1.100 | 1.200 |
| 04 | 49 | 2.000 ±.005 | 1.800 | 1.500 | 1.600 |
| 05 | 61 | 2.400 ±.005 | 2.200 | 1.900 | 2.000 |

| Inches | mm | Inches | mm | Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|---|---|---|---|
| .004 | 0.10 | .050 | 1.27 | .136 | 3.45 | .300 | 7.62 | 1.200 | 30.48 |
| .005 | 0.13 | .055 | 1.40 | .150 | 3.81 | .345 | 8.76 | 1.400 | 35.56 |
| .007 | 0.18 | .065 | 1.65 | .154 | 3.92 | .355 | 9.02 | 1.500 | 38.10 |
| .008 | 0.20 | .070 | 1.78 | .170 | 4.32 | .400 | 10.16 | 1.600 | 40.64 |
| .013 | 0.33 | .071 | 1.80 | .175 | 4.45 | .600 | 15.24 | 1.800 | 45.72 |
| .026 | 0.66 | .079 | 2.01 | .180 | 4.57 | .700 | 17.78 | 1.900 | 48.26 |
| .028 | 0.71 | .084 | 2.13 | .200 | 5.08 | .800 | 20.32 | 2.000 | 50.80 |
| .039 | 0.99 | .100 | 2.54 | .255 | 6.48 | 1.000 | 25.40 | 2.200 | 55.88 |
| .044 | 1.12 | .120 | 3.05 | .265 | 6.73 | 1.100 | 27.94 | 2.400 | 69.96 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerances are ±.005 (0.13 mm) on three place decimals and ±2° on angles.
4. These connectors mate with connectors specified in MIL-DTL-55302/7, and are primarily for use with multilayered printed wiring boards.
5. Positional tolerances of guide pins shall apply at datum plane X.
6. Pad(s) suitable for printed circuit board support are required. Dimensions and location(s) are optional.

FIGURE 1. Connectors, receptacle, .100 (2.54 mm) spacing - Continued.



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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1 and table I.

Material:
Guide sockets and guide bushings: Brass composition B, 60,000 to 70,000 psi tensile, as specified in ASTM B134/B134M or FC brass as specified in ASTM B16/B16M.

Plating:
Guide sockets and guide bushings: Gold over copper type II, code C, as specified in ASTM B488.

Contacts:
Gold in accordance with ASTM B488, type II, code C, class 1.27, over nickel plating in accordance with SAE-AMS-QQ-N-290, Class 2, 50 to 150 microinches.

Contact identification: Shall be numerical and sequential in pattern indicated.

Socket size: 23

Wire size: 22

Current rating: 5 amperes, maximum.

Oversize pin exclusion: Not applicable.

Mating and unmating: The maximum insertion force, in pounds shall not exceed a value equal to .5 times the number of contacts.

Contact engagement and separation force: The individual contact withdrawal force shall be .5 ounce minimum when tested with minimum diameter test pin SAE-AS31971-23X1.

Contact resistance: The average resistance of all contact pairs measured shall not exceed .010 ohm, and no individual contact pair shall have a resistance exceeding .020 ohm.

Probe damage: Test procedure EIA-364-25**.**
Contact size: 23
Test probe: .027 diameter.
Bending moment: .125 inch pounds.
Depth of insertion: .156 from connector face.

Dielectric withstanding voltage:
Sea level: 1000 volts rms, 60 Hz, ac.
High altitude: 500 volts rms, 60 Hz, ac.

Part or Identifying Number (PIN): M55302/111-(dash number from figure I).



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Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue, due to the extent of the changes.

Referenced documents. In addition to MIL-DTL-55302, this document references the following:

MIL-DTL-55302/7
ASTM B16/B16M
ASTM B134/B134M
ASTM B488
EIA-364-25
SAE-AMS-QQ-N-290

## CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-4448-000)

Review activities:
Army - AT, AV MI
Navy - AS, MC, OS
Air Force – 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.



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